

3 Datasets

3.11 Hourly City Load Factor

Source. Salam & El Hibaoui, Power Consumption of Tetouan City, UCI Machine Learning Repository, <https://archive.ics.uci.edu/dataset/849/power+consumption+of+tetouan+city>.

Name of the file. `hourly_load_factor.csv`

Short description. The original release records the active power consumed by the three distribution zones of Tetouan (Morocco) during 2017, sampled every ten minutes by the SCADA system of the local utility. The CSV we share is a derived, pre-cleaned version. For each (day, hour) of the year we have computed the hourly mean of the city-wide active power

$$P(d, h) = \text{Zone1}(d, h) + \text{Zone2}(d, h) + \text{Zone3}(d, h)$$

from the original 10-minute readings, and then the hourly load factor

$$LF(d, h) = \frac{P(d, h)}{\max_{h'} \bar{P}(d, h')},$$

i.e. the ratio between the average load in hour h and the highest hourly average of the same day. By construction $LF(d, h) \in [0, 1]$: the day-peak hour gets exactly 1 and the deepest off-peak hour gets the smallest value. From the full $365 \times 24 \approx 8700$ (day, hour) cells we have taken a uniform random subsample of 2000 rows and clamped the values to $(10^{-3}, 1 - 10^{-3})$ so that the Beta log-density is finite at the boundary.

The variables are:

- **date:** calendar date ($M/D/YYYY$);
- **hour:** hour of the day, integer in $\{0, \dots, 23\}$;
- **load factor:** hourly load factor as defined above, in $(10^{-3}, 1 - 10^{-3})$.

Task.

1. Fit a single Beta model and a Beta mixture with $H = 1, 2, 3, 4, 5$ components, performing model selection via WAIC.
2. Overlay the posterior predictive density on the empirical histogram and verify that the modes are recovered.
3. Post-hoc exploration. Assign each observation to its maximum-a-posteriori component, then plot the empirical fraction of observations in each component as a function of hour.
4. Advanced. Fit a second model which performs classification on hour using the component assignment as response variable, and plot the predicted probability of each component as a function of hour. Do the two plots match? Discuss the results. (Details for this step are presented in the additional details notebook.) The component assignment is predicted using a multinomial logistic regression with the Fourier-transformed hour as covariate, i.e. with predictors $\sin(k \cdot 2\pi h/24)$ and $\cos(k \cdot 2\pi h/24)$ to capture the periodicity of the day.